The elements of the second and third transition series resemble each other more closely than the elements of the first and second transition series. For example

Series	Element	Atomic radius
3d Series	Ti	132 pm
4d Series	Zr	145 pm
5d Series	Hf	144 pm

Actinoids:

The fourteen elements following actinium ,i.e., from thorium (Th) to lawrentium (Lr) are called actinoids. Unlike the lanthanoids, all the actinoids are radioactive and most of them have short half lives. Only thorium and uranium(U) occur in significant amount in nature and a trace amounts of Plutonium(Pu) is also found in Uranium ores.Neptunium(Np) and successive heavier elements are produced synthetically by the artificial transformation of naturally occuring elements by nuclear reactions.

Similar to lanthanoids, they are placed at the bottom of the periodic table.

Electronic configuration:

The electronic configuration of actinoids is not definite. The general valence shell electronic configuration of 5f elements is represented as $[Rn]5f^{0-14}6d^{0-2}7s^2$. The following table show the electronic configuration of actinoids.

Table : electronic configuration of actinoids

Name of the element	Atomic number	Symbol	Electronic configuration
Actinium	89	Ac	$[Rn]\ 5f^0\ 6d^1\ 7s^2$
Thorium	90	Th	$[Rn]\ 5f^0\ 6d^2\ 7s^2$
Protactinium	91	Pa	$[Rn]\ 5f^2\ 6d^1\ 7s^2$
Uranium	92	U	$[Rn]\ 5f^3\ 6d^1\ 7s^2$
Neptunium	93	Np	$[Rn]\ 5f^4\ 6d^1\ 7s^2$
Plutonium	94	Pu	$[Rn]\ 5f^6\ 6d^0\ 7s^2$
Americium	95	Am	$[Rn]\ 5f^7\ 6d^0\ 7s^2$
Curium	96	Cm	$[Rn]\ 5f^7\ 6d^1\ 7s^2$
Berkelium	97	Bk	$[Rn]\ 5f^9\ 6d^0\ 7s^2$
Californium	98	Cf	$[Rn]\ 5f^{10}\ 6d^0\ 7s^2$

XII U4-D-Block-Jerald Folder.indd 122 19-01-2021 12:18:52